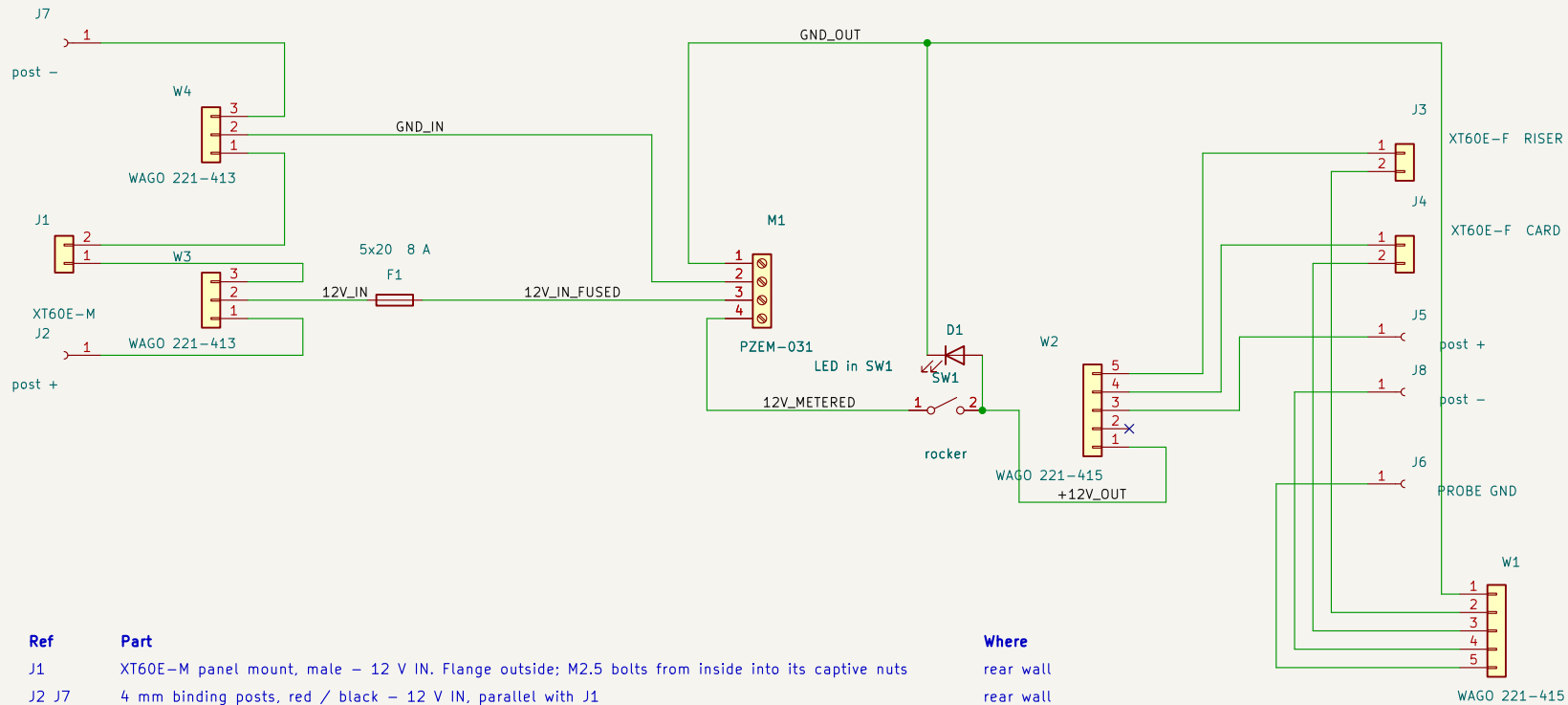


PCIe GPU test rig – power box wiring

Rev D 2026-09-30. 12V IN -> F1 -> M1 -> SW1 -> outputs. Every join is a WAGO 221 lever nut: W3/W4 on the input pair, W2 on the output +, W1 the ground bus.



Ref	Part	Where
J1	XT60E-M panel mount, male – 12 V IN. Flange outside; M2.5 bolts from inside into its captive nuts	rear wall
J2 J7	4 mm binding posts, red / black – 12 V IN, parallel with J1	rear wall
W3	WAGO 221-413 three-port: J1 +, J2, one wire to F1	inside
W4	WAGO 221-413 three-port: J1 –, J7, one wire to M1 terminal 2	inside
F1	5x20 mm panel fuse holder, 8 A fast-blow glass fuse	rear wall
M1	Peacefair PZEM-031 meter, 8-100 V / 0-20 A, built-in shunt	top deck
SW1	Amppper 20 mm illuminated rocker, 12 V 20 A, 3-pin	top deck
D1	the dot LED inside SW1 (yellow pin = LED –)	–
W2	WAGO 221-415 five-port: SW1 B in; RISER+, CARD+, OUT+ post out; one spare	inside
W1	WAGO 221-415 five-port, ground bus: M1 LOAD– in; RISER–, CARD–, OUT– post, PROBE out	base floor
J3	XT60E-F panel mount, female – OUT, RISER. Flange outside; M2.5 bolts + nuts	right wall
J4	XT60E-F panel mount, female – OUT, CARD (optional 2nd, parallel)	right wall
J5 J8	4 mm binding posts, red / black – OUT +/-	right wall
J6	4 mm binding post, black – PROBE GND (scope / DMM reference)	top deck

M1: the shunt is internal between 2 (DC IN–) and 1 (LOAD–). Input negative goes to 2 via W4, never to W1 – on the ground bus it bypasses the shunt and the meter reads 0 A.

SW1: A from M1.4, B to W2, yellow to ground (on the OUT – post terminal). Never swap B and yellow – dead short.

XT60E pigtails are 12 AWG, the top of the WAGO 221 range – push firmly. Runs 14 AWG. No solder in the box.

Supply: OWON at 12.0 V, OVP 12.6 V, 6 A current limit (the working max). F1 8 A fast 5x20. J4 is optional.

wiring/pcie_rig_power.kicad_sch – rev D 2026-09-30, edit in KiCad

lamMrCupp / pcie-gpu-test-rig

Sheet: /

File: pcie_rig_power.kicad_sch

Title: PCIe GPU test rig – power box wiring

Size: A4 Date: 2026-09-30

KiCad E.D.A. 10.0.6

Rev: D

Id: 1/1